

HELICS federation (ZeroMQ core), one broker and port per experiment

SSH: starts broker and federates

Grid federate
Python driver of GridPACK
pf_server on N MPI ranks

grid/meas

ns3/cmd_delivered

Network federate
NS-3: PMUs, links,
PDC, generator
nodes, attacker

ns3/pdc

cc/commands

Control-center federate
state estimation, bad data
detection, voltage control

ns3/status

grid/status

Observer federate
subscribes only,
writes live.json

cc/status

FastAPI service
queue, slot check, uploads,
run directories, results

HTTP

Node-RED dashboard
Experiments, Live, Inside,
Results, Cluster pages

HTTPS

Remote users
web browser over
VPN or SSH